



D3 Implementation

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D3b - Third Party Licences

The main third-party library we used was LibGDX which is a Java game development framework which we are using as the base to develop our game in Java so our game can be compiled and run on several different platforms like e.g. MacOS and Linux. The ability to develop a game for all our desires under a single code base allows us to save valuable time and effort converting and optimising our pre-existing code to run and perform on other platforms. LibGDX is provided under the Apache 2.0 licence, allowing us to use it for any commercial or non-commercial projects which is perfect for our situation. We are also able to edit the source code to change anything to how we desire and even redistribute our own modified or unmodified versions of the framework.

Some sprites like the WASD icons are originally from the Simple Keys Assets Pack made by beamedeighth available on itch.io. Our use case for these is allowed as the only directive for the fair use of these assets are *"use these assets in any kind of project you want, personal or commercial. Giving credits is not necessary but appreciated."* meaning we are allowed to use the sprites in the game with no problems, but we should give credit on the github page so we are using them in good faith. These sprites help us achieve NFR_LEGIBLE_TEXT through being large and clear in-game providing the player with an easily readable and understandable tutorial to the game which wouldn't have been possible if the images in the diagram were blurred and unreadable.

We also used several pieces of third party software in order to aid with the development of the game like for example Aseprite, a proprietary, source-available program. This, alongside photoshop, was mainly used for the development of sprites and assets used in the game. It requires a paid licence in order to be used but with that licence we can use it commercially for games, apps, or any creative project. However, unlike libGDX, we aren't allowed to redistribute the source code of these tools since they use commercial proprietary licenses. Use of pixel graphics supports one of our requirements of NFR_NO_GRAPHIC_CONTENT as pixel art is less detailed and so is generally safer for younger audiences even if the art is intended to appear unsuitable as the lack of detail hides anything we don't want. We also use it for NFR_FILE_SIZE as in our code we scale up pixel sprites, meaning we can store the sprites as very small images, keeping the file size down and letting us use this room elsewhere.

We have also added sound effects from mixkit.co. They are held under a 'Mixkit Sound Effects Free License' which means that they are free to use, to create an End Product that uses the sound effects as well as other things so that it is larger in scope and different to just the sound effect. This is well suited to use in our project, as the sound effects are included in the larger game and we are still free to use and distribute the game as we wish.

Implementation of Requirements

We have met all of the requirements stated in the final version apart from UR_MUSIC and FR_MUSIC_MUTE due to time constraints. However, these have a 'May' priority so we prioritised other requirements first, which is why music was left as not being able to be completed.